

HIERARCHICAL GENERALIZED LINEAR MODELS WITH R. APPLICATIONS IN PLANT/ANIMAL BREEDING AND GENETICS

Lars Rönnegård

Högskolan Dalarna

SLU

Sweden

Recap

- The estimates for an ordinary least squares problem is $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$
- The hat matrix $\mathbf{H}$ “puts the hat” on $\mathbf{y}$, i.e. $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$
- The *hat values* are the diagonal elements of $\mathbf{H}$.

Course outline

- **Linear regression**
 - diagnostic plots
- **Weighted least squares**
- **Generalized linear models**
- **Linear mixed models**
 - Estimating fixed and random effects
 - It's a weighted least squares problem!
 - Estimating variance components
 - Application on repeated observations
 - Fitting an animal model using the additive relationship matrix
 - Smoothing

Data example

Registered data

y	year
1.18	2001
3.28	2002
0.27	2002
3.69	2002
2.50	2002
5.53	2003
3.55	2003
3.24	2003
2.95	2004
5.29	2004
4.83	2004
3.94	2004
3.22	2004

Reported data

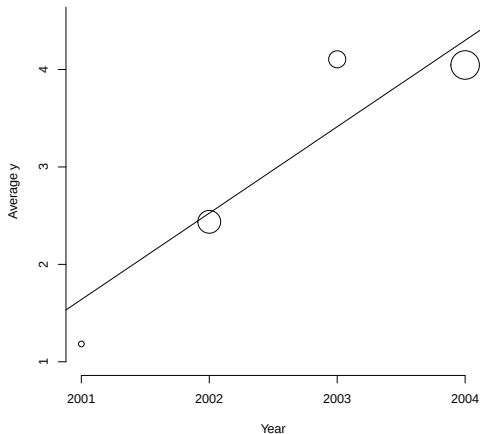
Average y	year	n
1.18	2001	1
2.44	2002	4
4.11	2003	3
4.05	2004	5

Regression

A weighted least squares example

Reported data

Average y	year	n
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Theory

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

$$\mathbf{e}_i \sim N(0, \frac{1}{w_i} \sigma_e^2)$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}$$

where $\mathbf{W}$ is a diagonal matrix with elements w_i .

R code

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Reported data

Average y	Year	n
1.18	2001	1
2.44	2002	4
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The following code give the same estimates and standard errors:

```
lm(y ~ year)
```

```
lm(y.average ~ Year, weights=n)
```

Theory

Estimates

$$\hat{\beta} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}$$

Variance matrix of estimates

$$V(\hat{\beta}) = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \sigma_e^2$$

Hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}$$

Data example

We have collected the following data 0.44, 0.77 and 2.56.

They were collected from a normal distribution with a variance of 1 and unknown mean, i.e $y_i \sim N(\mu, 1)$.

What is our best guess of μ ?

Plot the normal density function for different values of μ .

Data example

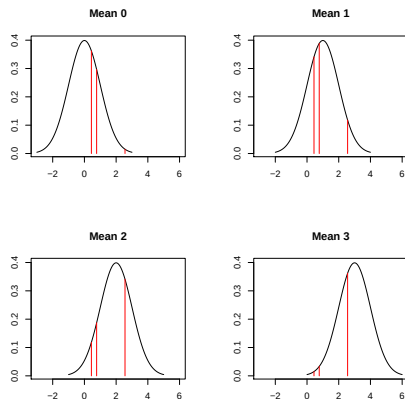
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Data example



Observed values are 0.44, 0.77 and 2.56.
What is our best guess of μ ?

Choose the density with highest vertical lines.

Likelihood computations

Likelihood value for each observation y_i as a function of μ

$$L_i(\mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y_i - \mu)^2\right)$$

Product of individual likelihoods

$$L(\mu) = \prod_{i=1}^n L_i(\mu)$$

We can plot $L(\mu)$ for different values of μ .

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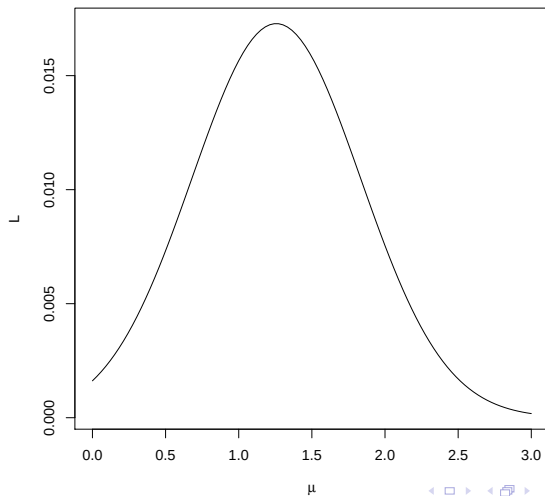
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Likelihood computations



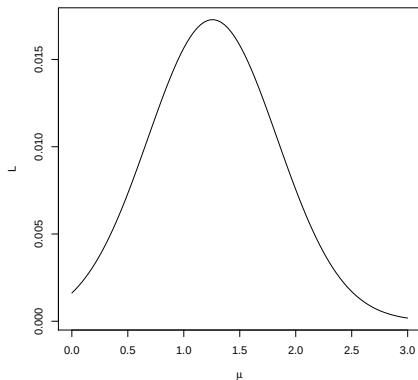
Maximum likelihood

$$L(\mu) = \prod \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}(y_i - \mu)^2)$$

Maximize $\ell(\mu) = \log L(\mu)$

$$\frac{d\ell}{d\mu} = 0$$

$$\hat{\mu} = \bar{y} = 1.26$$



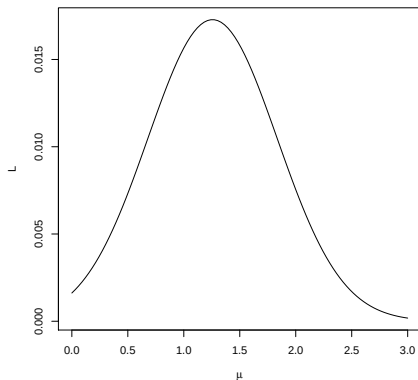
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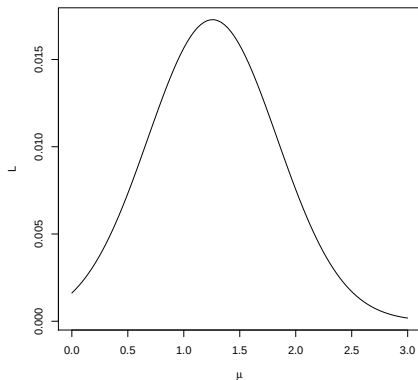
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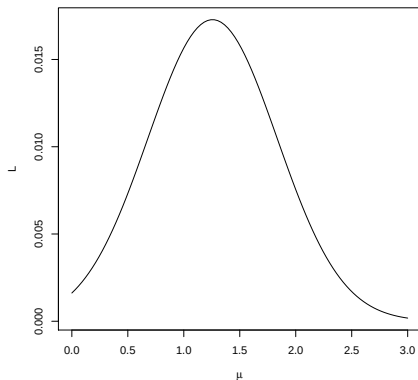
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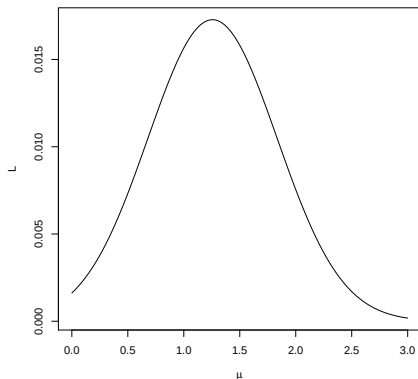
The standrd error of $\hat{\mu}$ depends on the curvature of $L(\mu)$

A sharp peak gives a small standad error.

A flat curve gives a large standad error.

$$SE = \sqrt{-\left(\frac{d^2\ell}{d\mu^2}\right)^{-1}}$$

$$SE = 0.66$$



Maximum likelihood

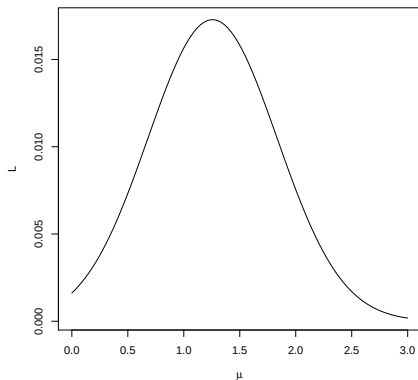
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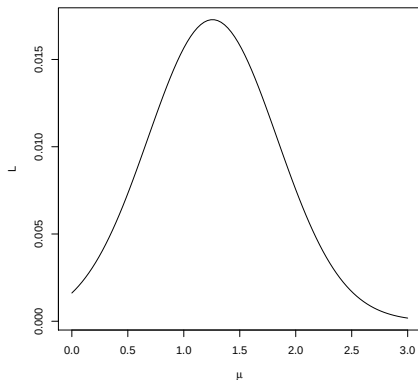
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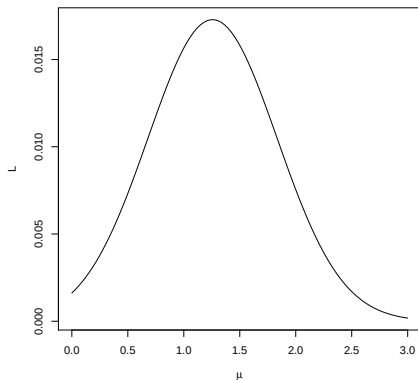
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Group exercise in pairs

We have collected the following data 1, 3, 3 and 4.

They were collected from a Poisson distribution with unknown λ , i.e $y_i \sim Po(\lambda)$.

The Poisson distribution has density $p(y) = \lambda^y \exp(-\lambda)/y!$

What is our best guess of λ ?

GLM

Poisson

$$L = \prod_{i=1}^n \frac{\mu^{y_i} e^{-\mu}}{y_i!}$$

$$\log(\mu) = X\beta$$

Binomial

$$L = \prod_{i=1}^n \binom{n}{y_i} \mu^{y_i} (1 - \mu)^{n - y_i}$$

$$\text{logit}(\mu) = X\beta$$

Gaussian

$$L = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(y_i - \mu)^2}$$

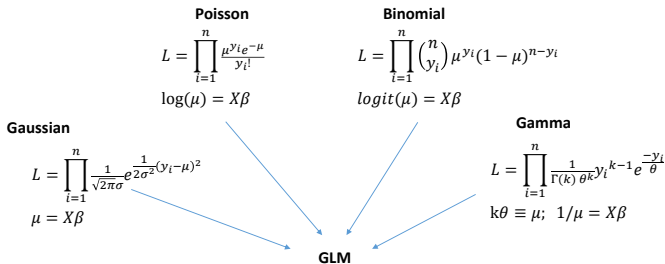
$$\mu = X\beta$$

Gamma

$$L = \prod_{i=1}^n \frac{1}{\Gamma(k)\theta^k} y_i^{k-1} e^{-\frac{y_i}{\theta}}$$

$$k\theta \equiv \mu; \quad 1/\mu = X\beta$$

GLM



Common estimation algorithm using iterative regression

- Fast and easy to implement
- ***Linear regression model checking tools!!!***

A method to solve the previous group exercise

We have collected the following data 1, 3, 3 and 4.

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The Poisson distribution has density $p(y) = \lambda^y \exp(-\lambda)/y!$

What is our best guess of λ ?

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y <- c(1, 3, 3, 4)
glm(y ~ 1, family=poisson(link=log))
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A method to solve the previous group exercise

```
y <- c(1, 3, 3, 4)
```

Call:

```
glm(formula = y ~ 1, family = poisson(link = log))
```

Deviance Residuals:

1	2	3	4
-1.2152	0.1485	0.1485	0.7054

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.0116	0.3015	3.355	0.000793 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 2.0185 on 3 degrees of freedom
Residual deviance: 2.0185 on 3 degrees of freedom
AIC: 15.268

Number of Fisher Scoring iterations: 4

GLM notation

We model the mean of the response $\mathbf{y}$ such that

$$E(y) = \mu$$

$$g(\mu) = \mathbf{X}\beta$$

In the Poisson example we have $\log(\mu) = \mathbf{X}\beta$

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Linearize the response

Linearize the response y , i.e $z = g(\mu) + (y - \mu)g'(\mu)$.

The variance of the linearized response z is: $\text{var}(z) = \text{var}[g(\mu) + (y - \mu)g'(\mu)]$.

Here y is the only stochastic part so:

$$\text{var}(z) = \text{var}[g(\mu) + (y - \mu)g'(\mu)] = \text{var}(y)(g'(\mu))^2.$$

In the Poisson case with a log link we have $\text{var}(y) = \mu$ and $g'(\mu) = \frac{1}{\mu}$.

So, $\text{var}(z) = \mu \left(\frac{1}{\mu}\right)^2 = \frac{1}{\mu}$

Consequently, if we fit the linearized response z with a linear model we need to use weights μ to make sure that the residuals have equal variance in this model. But μ depends on β so we need to iterate. The algorithm tends to converge quickly.

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Iterative weighted least squares

Algorithm for a Poisson distribution with log link

- Guess an initial value for β
- Compute $\mu = \exp(\mathbf{X}\beta)$
- Compute $z = g(\mu) + (y - \mu)g'(\mu)$ where $g'(\mu) = \frac{1}{\mu}$
- Let $w = \frac{1}{\mu}$ and fit the linear model $\text{lm}(z \sim x, \text{weights} = w)$
- Update β using the estimates from the linear model.
- Iterate between step 2 to 5 until convergence.

The algorithm tends to converge rather quickly.